Measurement Report 3.8.9.3

BenQ EX3210U @ 0, 0, 3840x2160 — 2026-11-08 17:57:05

Basic Information

Device:	BenQ EX3210U @ 0, 0, 3840x2160
Instrument:	SpyderX — LCD PFS Phosphor WLED, RGB LED
Correction:	None
Display profile:	EX3210U #1 2026-11-08 17-12 2.2 F-S XYZLUT+MTX
Display profile luminance:	121.2 cd/m²
Display profile whitepoint:	xy 0.3127 0.3284 (XYZ 95.2 100 109.27), CCT 6510K
Measured luminance:	120.8 cd/m²
Measured whitepoint:	xy 0.312 0.3282 (XYZ 95.06 100 109.67), CCT 6551K
Assumed target whitepoint:	6600K daylight, xy 0.3112 0.3276 (XYZ 94.99 100 110.26)
Measured black luminance:	0.118 cd/m²
Contrast:	1024:1
Testchart:	verify_extended.ti1
Simulation profile:	None
Gamma mapping:	N/A
Whitepoint simulation:	N/A
Chromatic adaptation:	Bradford
Devicelink profile:	None
Evaluation criteria:	RGB
Date:	2026-11-08 17:57:05

Summary

Criteria	Nominal	Recommended	#	Actual	Result
Measured vs. assumed target whitepoint ΔE*00	<= 2	<= 1		0.4	OK ✓✓
Measured vs. display profile whitepoint ΔE*00		<= 1		0.42	
Average ΔE*00	<= 1.5	<= 1		0.31	OK ✓✓
Maximum ΔE*00	<= 4	<= 3	02	0.96	OK ✓✓

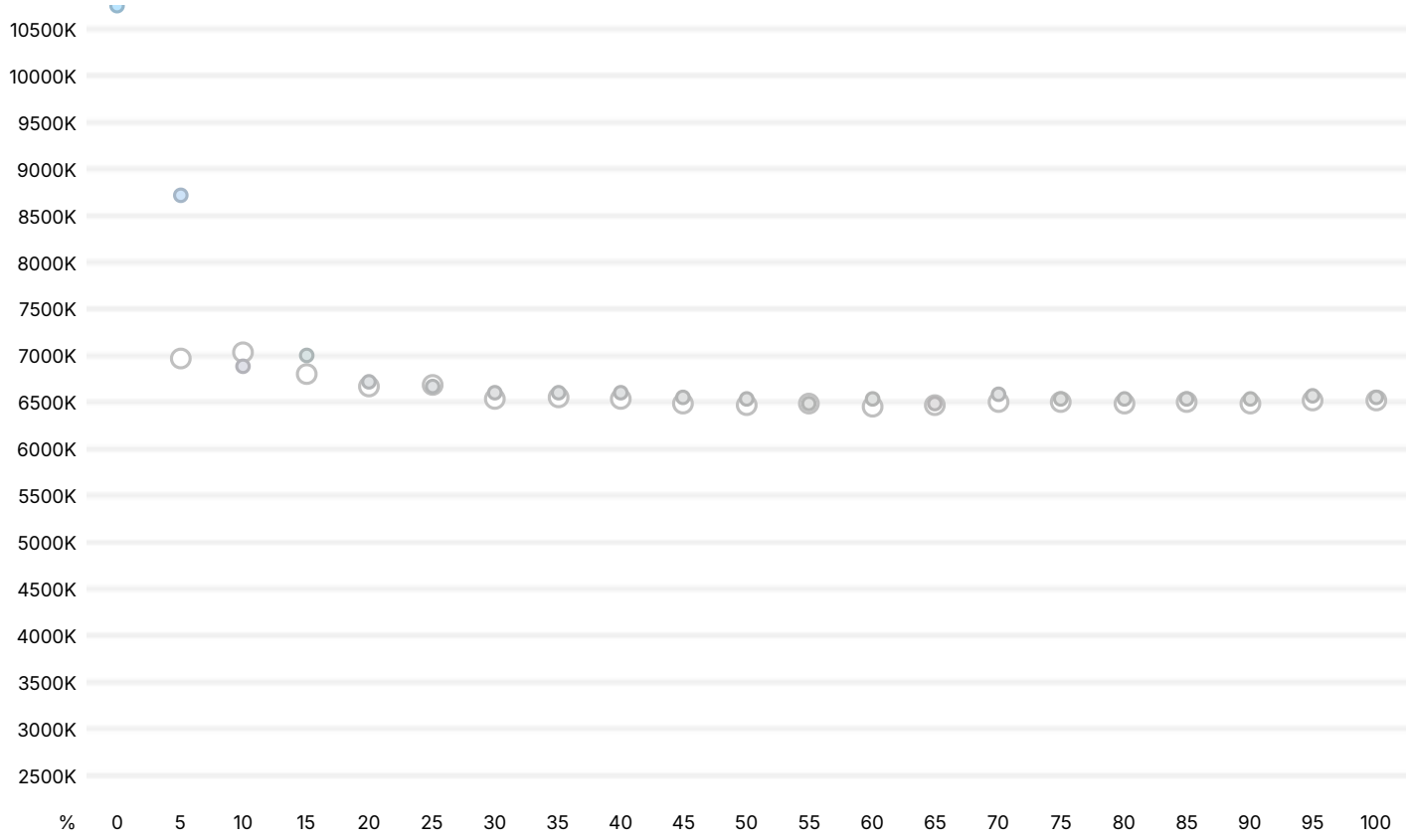
✓ Nominal tolerance passed
✓ Recommended tolerance passed

Overview

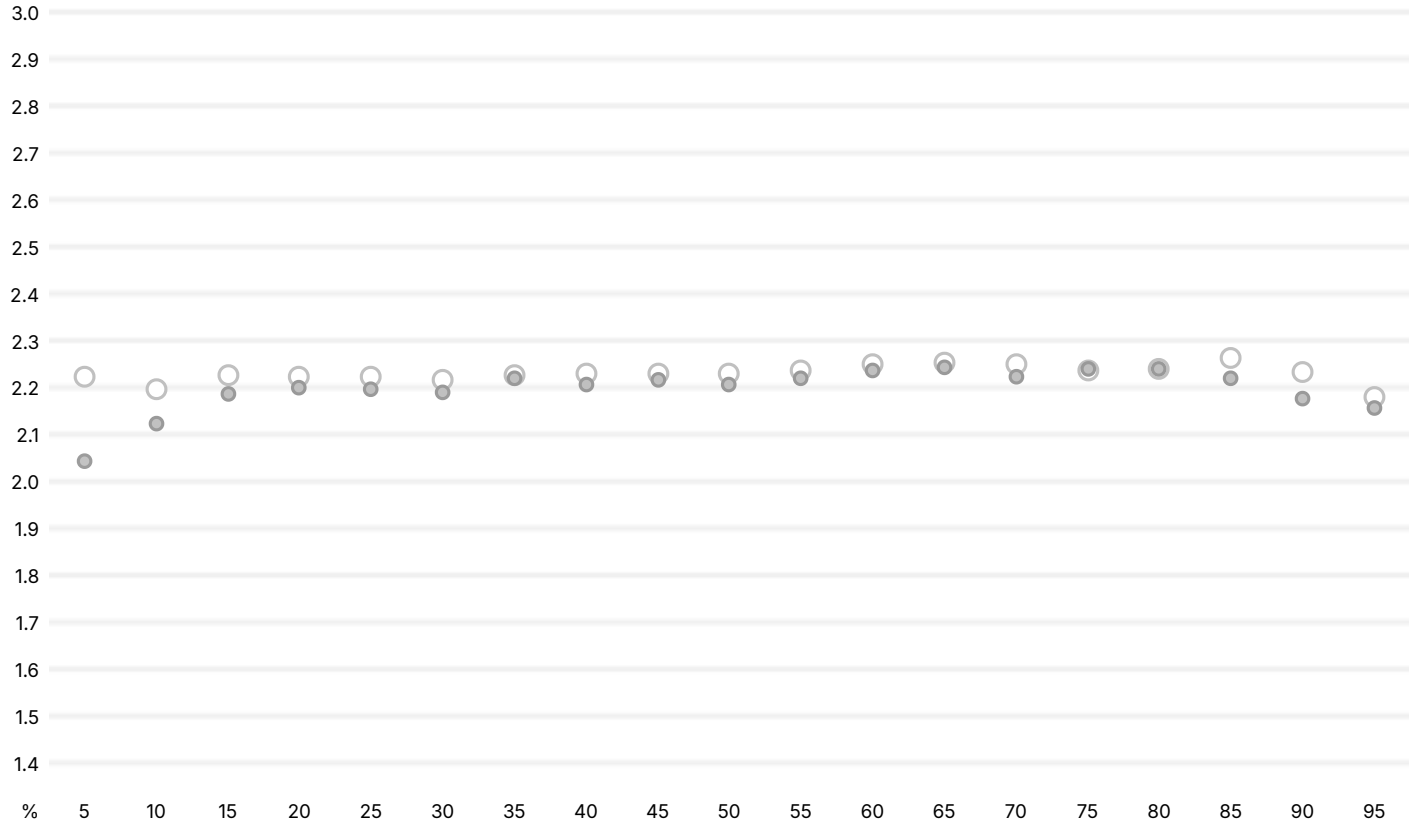
#	Device Values			Nominal Values					Measured Values				Color distance			
	R	G	B	L*	a*	b*	γ		L*	a*	b*	γ	$\Delta L^*_{00/w}$	$\Delta C'_{00/w}$	$\Delta H'_{00/w}$	ΔE_{00}
01	255	255	255	100	0	0			100	0	0		0	-0.01	0	0.01
02	0	0	0	0	0	0			0.87	-0.15	-0.8		0.5	0.82	0	0.96
03	13	13	13	1.15	0.1	-0.2	2.22		1.97	-0.23	-0.74	2.04	0.48	0.56	-0.47	0.87
04	26	26	26	5.73	-0.06	-0.71	2.2		6.82	0.13	-0.66	2.12	0.66	-0.02	0.29	0.72
05	38	38	38	12.4	-0.59	-0.33	2.23		13.13	-1.03	-0.43	2.19	0.47	0.62	-0.11	0.78
06	51	51	51	19.17	-0.07	-0.44	2.22		19.63	-0.19	-0.37	2.2	0.32	0.01	-0.2	0.37
07	64	64	64	25.51	-0.4	-0.37	2.22		26.01	-0.16	-0.3	2.2	0.37	-0.31	0.17	0.51
08	77	77	77	31.66	-0.48	0.2	2.22		32.15	-0.04	-0.18	2.19	0.39	-0.54	0.52	0.84
09	89	89	89	37.24	-0.25	0.01	2.23		37.32	-0.26	-0.08	2.22	0.07	0.04	0.09	0.12
10	102	102	102	42.71	-0.26	0.03	2.23		43.11	-0.26	-0.12	2.21	0.37	0.02	0.14	0.4
11	115	115	115	48.09	-0.12	0.18	2.23		48.3	-0.2	0.16	2.22	0.21	0.08	0.09	0.24
12	128	128	128	53.3	-0.12	0.31	2.23		53.66	-0.23	0.22	2.21	0.35	0.05	0.18	0.39
13	140	140	140	58.29	-0.33	0.39	2.24		58.51	-0.43	0.7	2.22	0.2	0.32	-0.12	0.39
14	153	153	153	63.1	-0.2	0.49	2.25		63.27	-0.59	0.5	2.24	0.15	0.42	0.38	0.58
15	166	166	166	67.95	0.15	0.23	2.25		68.06	0.57	0.08	2.24	0.08	0.53	-0.35	0.64
16	179	179	179	72.76	-0.02	0.05	2.25		73.05	-0.07	-0.16	2.22	0.21	0.13	0.18	0.31
17	191	191	191	77.6	-0.06	0.16	2.24		77.58	-0.53	0.53	2.24	-0.02	0.75	0.19	0.78
18	204	204	204	82.2	-0.53	0.56	2.24		82.19	-0.4	0.45	2.24	0	-0.22	-0.02	0.22
19	217	217	217	86.62	-0.32	0.26	2.26		86.85	-0.26	0.35	2.22	0.15	-0.02	-0.13	0.2
20	230	230	230	91.24	-0.34	0.42	2.23		91.46	-0.54	0.57	2.18	0.13	0.33	0.06	0.36
21	242	242	242	95.75	-0.16	0.03	2.18		95.8	-0.11	-0.01	2.16	0.03	-0.07	0.03	0.08
22	128	0	0	33.29	75.08	55.2			33.6	74.97	54.23		0.25	-0.13	-0.35	0.45
23	255	0	0	58.69	113.55	95.43			58.57	113.52	95.57		-0.11	0.01	0.05	0.12
24	255	128	128	69.55	74.99	30.65			69.5	74.54	30.27		-0.04	-0.12	-0.09	0.16
25	0	128	0	49.21	-96.09	57.98			49.61	-95.73	57.61		0.41	-0.08	0.04	0.42
26	0	255	0	85.43	-149.37	90.56			85.52	-149.1	90.65		0.06	-0.02	-0.05	0.08
27	128	255	128	87.58	-97.46	52.14			87.44	-95.84	51.97		-0.09	-0.25	-0.19	0.33
28	0	0	128	15.42	46.39	-77.35			15.87	46	-77.16		0.3	-0.07	-0.15	0.33
29	0	0	255	30.63	69.8	-115.49			30.5	69.71	-115.27		-0.1	-0.03	0.02	0.11

30	128	128	255	57.58	27.51	-67.95		57.81	26.99	-67.38	0.21	-0.17	-0.2	0.27
31	0	128	128	50.51	-64.09	-11.56		50.65	-63.48	-11.79	0.14	-0.14	0.17	0.26
32	0	255	255	85.74	-94.61	-21.79		85.76	-94.61	-21.75	0.01	0	-0.02	0.02
33	170	255	255	91.59	-45.6	-10.68		91.74	-45.4	-10.53	0.09	-0.07	-0.06	0.13
34	128	0	128	36.09	80.06	-37.62		36.46	79.31	-37.38	0.31	-0.16	-0.03	0.35
35	255	0	255	64.13	121.56	-57.59		64.04	121.85	-57.65	-0.08	0.04	0.02	0.09
36	255	170	255	80.53	55.63	-27.28		80.46	56.02	-27.38	-0.04	0.11	0.04	0.12
37	128	128	0	53.15	-10.07	66.7		53.44	-10.33	66.43	0.28	-0.06	0.17	0.33
38	255	255	0	94.42	-14.8	107.14		94.48	-14.81	107.6	0.03	0.08	-0.02	0.09
39	255	255	170	97.09	-9.33	42.27		97.25	-9.47	42.3	0.1	0.02	0.09	0.13
40	170	85	85	48.54	63.1	27.46		48.67	63.88	27.97	0.13	0.22	0.08	0.27
41	85	170	85	62.24	-82.77	45.23		62.16	-82.52	45.59	-0.07	-0.01	-0.15	0.16
42	85	85	170	39.08	24.43	-58.05		39.26	23.82	-57.85	0.16	-0.11	-0.36	0.35
43	85	170	170	63.35	-58.86	-11.65		63.33	-58.94	-11.68	-0.01	0.02	0.01	0.03
44	170	85	170	50.5	71.04	-33.49		50.64	71.42	-33.35	0.15	0.06	0.11	0.19
45	170	170	85	67.98	-10.26	57.93		68.12	-10.02	58.38	0.11	0.11	-0.19	0.24
46	255	0	170	61.29	117.35	-18.47		61.21	117.58	-18.54	-0.07	0.04	-0.01	0.08
47	170	255	0	89.31	-73.19	97.8		89.43	-72.39	98.1	0.08	-0.04	-0.26	0.27
48	0	170	255	66.15	-43.28	-55.8		66.13	-43.03	-55.98	-0.01	0	0.12	0.12
49	0	255	170	85.19	-123.6	22.43		85.18	-123.14	22.1	0	-0.08	0.08	0.11
50	170	0	255	51.31	101.72	-79.63		51.15	102.16	-79.78	-0.15	0.07	0.05	0.17
51	255	170	0	77.33	42.21	96.6		77.33	42.27	96.88	0	0.05	0.03	0.06

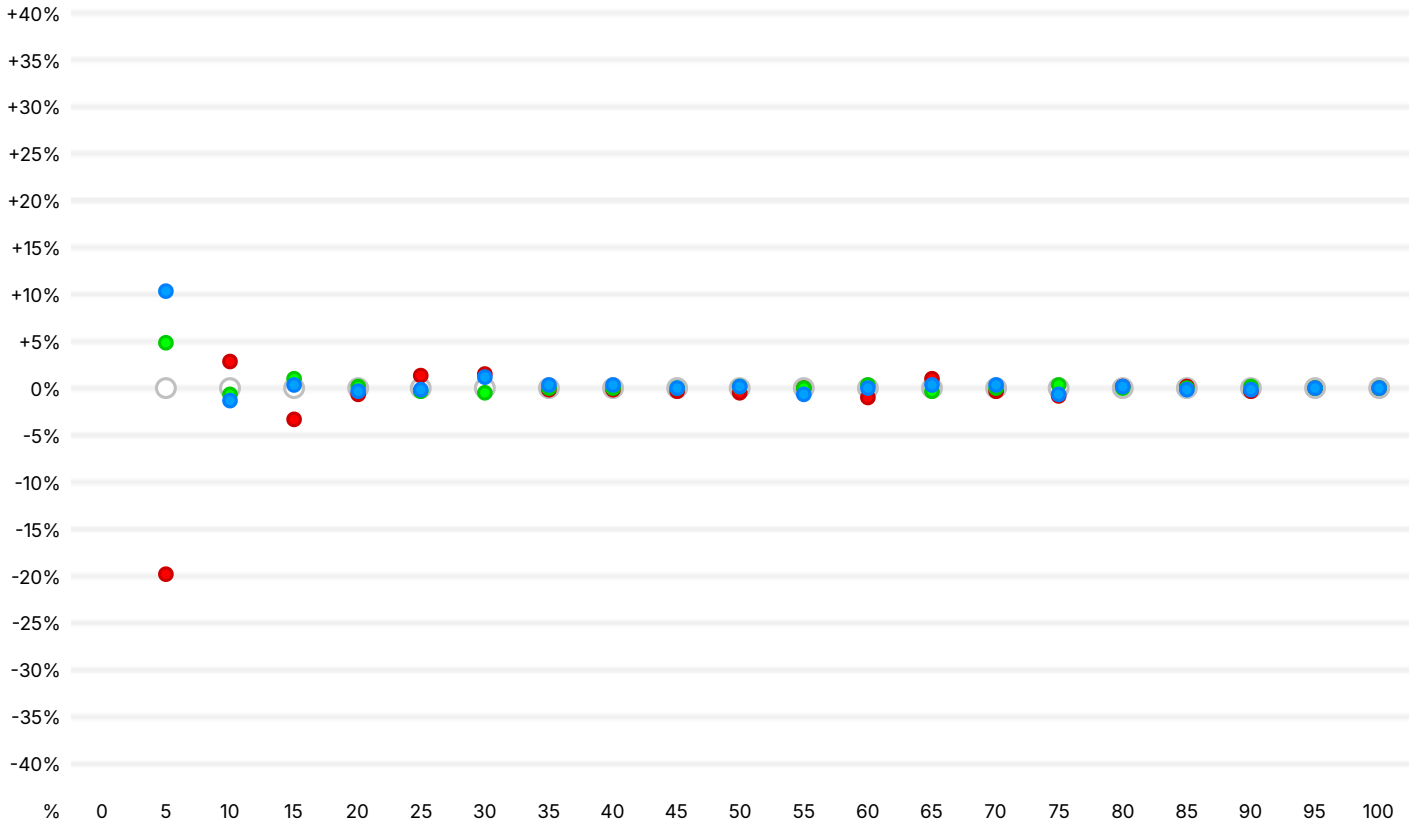
Correlated Color Temperature



Gamma



RGB Gray Balance



Gamut a*b*

